Model-Free RL for the Fully Cooperative Consensus of Multi-Agent Systems

Reinforcement Learning, Course Project (with Kimia Aghili)

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Problem

We reproduced H. Wang and M. Li, "Model-free reinforcement learning for fully cooperative consensus problem of nonlinear multiagent systems," *IEEE TNNLS*, 2020. N agents $\dot{x}_i = f(x_i) + g(x_i)u_i$ communicate over a directed graph with weights e_{ij}, in-degrees d_i and Laplacian L. With the local neighbourhood errors $\delta_i = \sum_j e_{ij}(x_i - x_j)$, i.e. $\delta = (L \otimes I_n)x$, all agents minimise one shared cost

$$J = \int_0^\infty \tfrac{1}{2}Q(\delta) + \tfrac{1}{2}\textstyle\sum_i u_i^\top R_i u_i \, dt.$$

Minimising the Hamiltonian gives $u_i^* = -d_i R_i^{-1} g_i^\top \partial V^*/\partial \delta_i + \sum_j e_{ji} R_i^{-1} g_i^\top \partial V^*/\partial \delta_j$: agent i needs its own error and the errors of the agents that listen to it.

Off-policy integral Bellman equation

Data are generated once with an exploring input u. For target policies u_i^k, the paper's eq. (20) is

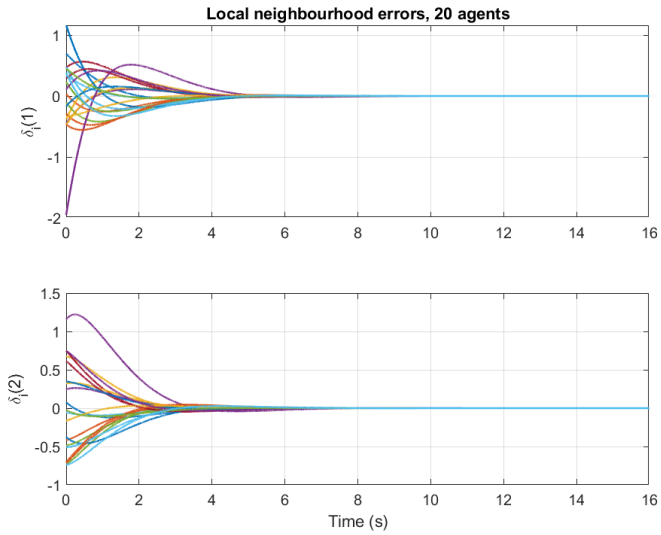
$$V^{k+1}(\delta(t)) - V^{k+1}(\delta(t{+}T)) = \int_t^{t+T} \Big[\tfrac{1}{2}Q(\delta) + \tfrac{1}{2}\sum_i (u_i^k)^\top R_i u_i^k\Big] d\tau + \int_t^{t+T} \sum_i (u_i^{k+1})^\top R_i (u_i - u_i^k)\, d\tau,$$

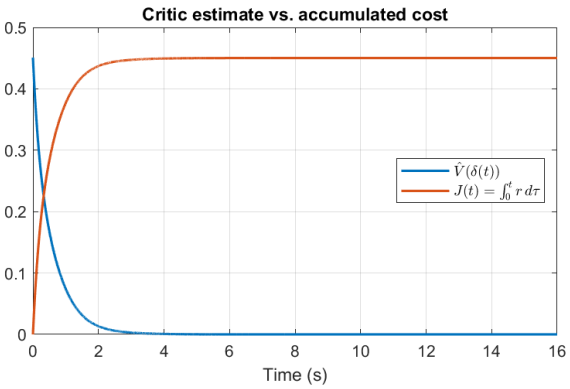
which holds without knowing f or g. With $V^{k+1} = W_c^\top \phi(\delta)$ and $u_i^{k+1} = W_{ai}^\top \psi_i$ it is linear in all weights. I solve it by least squares over 3,600 intervals of $T = 0.05\,\mathrm{s}$ at each policy-iteration step (Algorithm 3), starting from the admissible $u^0 = 0$. Because δ lies in a subspace (the rows of L are dependent), some basis columns are collinear and the minimum-norm solution is used.

Results

Example	consensus	$\hat{V}(\delta(0))$	J	gap
Linear check (5 agents)	6.2 s	1.3398	1.3245	1.2%
5 nonlinear agents (paper V-A.1)	6.5 s	0.4509	0.4501	0.18%
20 nonlinear agents (paper V-A.2)	7.5 s	5.3070	5.2529	1.0%

The paper reports consensus after about 6 s and 8 s with gaps of 0.83% and 0.29%. In the linear check the centralised Riccati equation gives $V^* = 1.3233$, so the learned distributed controller is within 0.1% of the best centralised one; the 1% critic gap comes from the distributed actor structure. In both nonlinear examples the agents are stable on their own and R_i is large (10 and 200), so the optimal input is small and J is close to the cost with $u = 0$ (0.4520 and 5.2573).





Choices not fixed by the paper

The paper gives neither initial states nor activation functions. I used quadratic monomials of δ for the critic (plus quartic terms when Q is quartic), each agent's error and its listeners' errors for the actor, plus a copy scaled by $\sin x_{i1}$, the agent's own input gain. The exploring input is a sum of six sinusoids per agent.